

Colan Biemer

Experience

Post-Doc | Drexel University | 2026 - Present

- Rebuilding the Cira Centre game engine as a modern, language-agnostic asyncio-based system using websockets and aiohttp, so games or visualizations for the building can be written in any language.
- Advised undergraduate and master's students on the [Cira Centre Games Project](#) at Drexel University and New Jersey Institute of Technology.
- [Successfully ran four student-designed games](#) displayed on the Cira Centre skyscraper's LED array during Philly Tech Week 2026, teaching middle schoolers software development.

PhD Researcher | Northeastern University | 2019 - 2025

- Published [eight peer-reviewed papers](#) on procedural generation and dynamic difficulty adjustment at leading game conferences, including a Best Paper nomination at IEEE CoG 2022.
- Engineered a full research pipeline for online player studies, utilized by studies with three custom-made browser-based games, including a [custom Go server](#) for block-randomized condition assignment and data collection.
- Created a content generation system in Python and TypeScript that dynamically adjusts difficulty in real-time using Markov Decision Processes.
- TA for 14 courses in game engine development, graphics, and C++; nominated for the Outstanding Graduate Teaching Award.

Applications Programmer | Brain Game Center | 2017 - 2019

- Lead developer of [PolyRules!](#), a Unity-based application designed to improve cognitive [task-switching](#). Used in a [published weight loss study](#) where participants showed 0.3 greater BMI reduction than the control group.
- Created a [shared submodule](#) to consolidate tools and core features across all applications, eliminating code duplication across repositories.
- Built a research data platform with AWS Cognito, Lambda, and S3 enabling researchers to configure studies (A/B conditions, app parameters), manage participants, and collect data.

Information Technology Research Center Engineer | BMW | 2016

- Reduced [Hive SQL](#) query runtime from one month to ~6 hours by batching the query into pages and processing updates in parallel across a 50-node cluster.
- Developer and administrator for an [ELK](#) stack and cluster to provide [NHTSA](#) data for analysts.

- Built a custom eight-node Raspberry Pi cluster and designed a hands-on training program around it, teaching cluster computing to interns and new hires in a low-stakes environment.

R&D Innovations Engineer | IPipeline | 2015

- Designed and built "Text-a-Quote," a custom Node.js SMS chatbot using Twilio and MongoDB, driven by a finite-state machine that collected user data and delivered life insurance quotes through a proprietary internal API.
- Shipped "Pipe-SMS," a rate-limited Node.js and Twilio API abstracting transactional SMS (passwords, verification codes) for customer-facing products.

Other Projects

- Developer and maintainer of [adjust.h](#), a single-header C11 library for live parameter adjustment without re-compilation, with 260+ GitHub stars.
- Building a billiards puzzle game with C11 and SDL3, including custom 2D rigid body physics, an in-game level editor, and a progression management system.
- Developing [staunch](#), a C11 utility library covering dynamic arrays, linear algebra, geometry, physics, sorting, and more, built alongside the billiards game as a practical, project-driven foundation.
- Producing educational C and Raylib programming tutorials on YouTube, growing a channel to ~500 subscribers with individual videos reaching 18k+ views.
- Developed a [dual-sided Tetris game](#) for the Cira Centre skyscraper using a custom game engine, with SSH and named pipes for server communication. Earned the Guinness World Record for World's Largest Architectural Video Game.
- Created a [graph reduction algorithm](#) that shrinks procedural maze graphs by 31% of nodes and 65% of edges, speeding up pathfinding algorithms like A*.

Skills

Main Languages: C, Python, TypeScript

Other Languages: C#, C++, Go, Java, Rust, OCaml

Infrastructure/Tools: AWS, GCP, SQL, SQLite, MongoDB, Git, Bash, Docker, Hive, Hadoop, and more

Education

Northeastern University, PhD, Computer Science, 2025

Northeastern University, MS Computer Science, 2021

Drexel University, BS, Computer Science (w/ Math Minor), 2017